

“I Could Wear It All of the Time, Just Like My Wedding Ring:” Insights into Older People’s Perceptions of Smart Rings

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Figure 1: Photographs from our exploratory study conducted to collect older people’s perceptions of and preferences for smart ring technology. Our participants tried out three commercial rings (Zero, Litho, and Omni), representative of *rings*, *ring-like*, and *ring-ready* devices, rated their user experience, and expressed preferences for using smart rings in their everyday life.

ABSTRACT

We present findings from an examination of older adults’ perceptions of and preferences for smart rings, a unique class of interactive wearables gaining traction in the consumer market. To this end, we conducted semi-structured interviews with nine older adults, aged between 61 and 69 years old, to understand their perceptions regarding several smart ring models representative of *rings*, *ring-like*,

and *ring-ready* finger-augmentation devices. We also elicited our participants’ preferences for using smart rings in their everyday life, such as for controlling various home appliances and electronic devices. Our results show an overall positive experience and openness toward adopting smart rings, which were perceived useful to assist with various activities. Based on our findings and following the aging research agenda in HCI, we propose future work opportunities for designing smart ring technology tailored to the needs and preferences of older adults.

CCS CONCEPTS

• **Human-centered computing** → **User studies.**

KEYWORDS

Smart rings, older adults, study, interviews, user experience

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1 INTRODUCTION

Smart rings are a distinctive category of personal devices in the landscape of wearables designed for finger augmentation [34] with their unique form factor, particular input and output modalities, and specific application opportunities [41]. Many commercial smart ring models are available on the market, providing technical solutions for notifications delivery, security, payment, health and wellness, and data transfer. For example, Zero¹ can be used to control devices with mid-air gesture input, Litho² enables intuitive interaction with virtual objects in AR/VR applications via pointing and touch-based input, and Omni³ is a NFC ring that stores and shares information securely. Due to the convenience, always-availability, lightweight form factors, and diversity of applications enabled by smart rings to support everyday life activities, their global market is expected to increase, as forecast by industry reports [24,36].

In this context, understanding the user experience of wearing and using a smart ring is key to the design of intuitive ring-based interactions and development of useful, effective applications that meet users' needs. However, most of the scientific research on smart rings has focused on introducing new technology, e.g., new sensing techniques [6,22,35], gesture recognition algorithms [9,43,48], and feedback delivery modalities [17,31,50] for smart rings, and less on understanding users' perceptions, preferences, and overall experience with this unique category of a wearable device. Regarding the latter, scientific examinations have primarily addressed young adults or people in their early middle ages, e.g., participants aged between 21 and 45 years old in Gheran *et al.*'s [13,15] end-user elicitation studies that documented end-users' preferences for smart ring gestures, and young adults between 20 and 38 years old in Bardot *et al.*'s [3] in-situ diary study of smart ring usage. This state of things has led to a limited understanding of users' needs and preferences for smart ring technology, mostly representative of the experience of an end-user audience composed of young adults.

To the best of our knowledge, the user experience of smart rings has not been examined for older people, despite the importance of understanding the relationship between technology and growing old [42]. There is evidence in this regard from analyzing the eighty-four scientific references of the GestuRING [41] database and systematic literature review, where we found only seven papers that included people over 60 years old in user studies about smart rings, but merely as regular participants in mixed-age samples, without addressing their specific needs or preferences for smart rings as a distinct user group. Unfortunately, population aging comes with implications for how new technology is used and how it should be designed in the first place [42]. To address this gap in design knowledge about smart rings tailored to the needs of older

people, we provide the first empirical examination of older people's perceptions of and preferences for smart ring technology with semi-structured interviews conducted with nine participants aged between 61 and 69 years old. We report an overall positive user experience with three commercial smart ring models, representative of *rings*, *ring-like*, and *ring-ready* devices [41], moderate perceived usability, usefulness, and ease of use—nevertheless, a result that places smart rings on a comparable level with smartphones,—and insights into the preferred form factor for smart rings, the fingers on which to wear them, and categories of electronic devices and home appliances to control with a smart ring, respectively.

2 RELATED WORK

We relate to prior work on smart rings and interactive systems for older people, at the intersection of which lies the scope of our work.

Research on smart ring technology has received growing interest in the recent years due to the many desirable qualities of smart rings, which make them an optimal wearable device for implementing ubiquitous interactions. For example, Wolf *et al.* [45] proposed a micro-gesture user interface for smartphone input implemented with two smart rings with inertial sensing, Bardot *et al.* [2] introduced ARO, a smart ring prototype designed for in-air, on-ring, and on-object input while grasping objects, and Bianchi and Je [4] proposed combined ring and touch input for enhancing interactions with touchscreen devices. Researchers have also examined interaction techniques that exploit the quality properties of smart rings, such as gesture input performed on the surface of uninstrumented objects [49], haptic output via poking [17], proximity-based NFC interactions [5], and on-finger controls for attention demanding tasks [14], to mention a few. Vatavu [40] introduced a taxonomy of gestures captured with devices worn on the index finger, including rings, and demonstrated several quality properties of index-finger augmented devices and gestures. Gheran *et al.* [13] reported empirical results from the first end-user gesture elicitation study conducted with smart rings, and proposed design implications for ring-based gesture input reflective of user behavior, such as reusing conventional touch gestures from smartphones for on-ring input due to their familiarity to smartphone users. However, this prior research has not addressed older people, for which we could not find any specific research questions or examinations; see Vatavu and Bilius' [41] systematic literature review of ring gesture input.

Previous research has reported that older people, aged +60 years, have specific preferences and needs when interacting with mobile and wearable technology [12,27,32,46,47]. Thus, understanding wearable interactions for older people represents an active topic of scientific investigation [30]. In particular, Ferron *et al.* [12] found that accessible gesture input fosters perception of natural interactions and ease of use, and improves memorization. An implication for touchscreen input is user interfaces with touch buttons of custom size and adaptability to less precise touch target acquisition due to aging [28]. Besides gestures, speech input has also been considered for older people to identify aspects of technology acceptance, e.g., Wulf *et al.* [46] and Schlögl *et al.* [32] reported that speech-based interactions were found pleasant and suitable for everyday use. In this context, much of the work from the scientific literature about wearables and older people has primarily focused on

¹<https://web.archive.org/web/20170309191236/http://ringzero.logbar.jp>

²<https://www.litho.cc>

³<https://store.nfcring.com/products/omni>

Table 1: Technical characteristics of the smart ring devices used in our study; see Figure 1 for photographs.

Device	Material	Weight (g)	Thickness (mm)	Width (mm)	Applications	Input	Output	Category [†]
Zero ¹	polycarbonate	5	6.7	13.2	control devices	touch button, motion sensing	LED, vibrations	ring
Litho ²	zinc die-cast chassis, back-sprayed glass	22	4.8	18.3	control devices, interact in AR/VR	capacitive trackpad, motion sensing	LED, vibrations	ring-like
Omni ³	scratch resistant zirconia ceramic	4	2.9	6.8	unlock smartphone, share data, access control, make payments	proximity (NFC)	–	ring-ready

[†]According to Vatavu and Bilius [41], *rings* are finger-worn devices with a ring form factor; *ring-like* devices are worn on the finger, yet they deviate from the form factor of a ring, but enable similar input as rings; *ring-ready* devices are multiform, multi-purpose devices that, among several form factors, can also take the form of a ring (p. 715).

applications in health and well-being, e.g., fall detection [19], blood pressure prediction [16], or step counting [25]. However, the needs and preferences of older people for smart ring technology have not been examined, which limits the current understanding on designing effective ring-based interactions and applications. Next, we present our study conducted to gain insight into such preferences.

3 STUDY

We conducted a study to gain an understanding of older people’s perceptions of smart ring technology, and elicit what they expect and look forward to in terms of practical uses of smart rings in their everyday life.

3.1 Participants

Nine older adults (five male and four female), aged between 61 and 69 years old ($M=64.9$, $SD=2.8$ years), were recruited via convenience sampling. At the time of the study, all but one of them had retired. Participants’ past occupations included educator, accountant, researcher, truck driver, tailor, mechanic, glazier, and energy specialist. None of the participants had prior experience with smart rings before our study, and Figure 2a gives an overview of their self-reported experience with and use of other smart devices.

3.2 Apparatus

We used three commercial smart rings—Zero,¹ Litho,² and Omni³ (see Figure 1),—from the smart ring models available on the consumer market, which we used to introduce our participants to a category of smart device they had never tried out before our study. These smart ring models can be used for controlling and interacting with applications on personal and public devices, storing, sharing, and transferring information, and streamlining access control and POS payments, respectively. Also, these three devices are convenient for eliciting feedback in our exploratory study, since they have different form factors, materials, and weights; see Table 1 for the technical details of the three ring device models used in our study. Furthermore, they are representative of the three categories of smart rings—*rings*, *ring-like*, and *ring-ready* devices—according to Vatavu and Billius’ [41] taxonomy. For example, Litho is *ring-like* due to its different form factor compared to a ring, yet it still enables the same kinds of interactions. Omni is an example of a *ring-ready* device because of its versatility as it can be worn in a multitude of ways, e.g., attached to a necklace, instead of just as a ring.

3.3 Procedure

We conducted the interviews at the participants’ homes. Each of the participants filled out a questionnaire with demographic information, including their previous experience with various electronic and computer devices, such as smartphone, smart TV, tablet, laptop, smartwatch, etc. We then introduced the three smart rings, and described the applications they could be used for, a stage during which we also used a few online videos⁴ to demonstrate specific use cases for smart rings, e.g., POS payments. The participants tried out the rings to experience the different form factors, weights, and materials; see Figure 1 for several photographs taken during our study. We continued with one-to-one semi-structured interviews to collect aspects of perception, needs, and desires for smart ring technology, representing the measures of our study; see Subsection 3.4 for the corresponding questions from our interviews. We recorded participants’ answers using a Google Forms document, which was convenient to structure the questions to make sure they were presented identically to all of the participants. The study lasted between 25 and 30 minutes per participant.

3.4 Measures

By following prior work that examined the user experience of wearables and finger-augmentation devices [37,38], we collected the following measures of user experience with smart rings using 7-point Likert scale statements with items ranging from 1 (“strongly disagree”) to 7 (“strongly agree”):

- PERCEIVED-EFFICIENCY, in response to the statement “Interacting with a smart ring is efficient.”
- PERCEIVED-NATURALNESS, in response to the statement “Interacting with a smart ring is natural.”
- PERCEIVED-COMFORTABILITY, in response to the statement “Smart rings are comfortable to wear.”
- UNWANTED-ATTENTION, in response to the statement “Using a smart ring in the presence of other people would draw unwanted attention to myself.”

Furthermore, to elicit and understand preferences for practical uses of smart rings, we measured:

- PREFERRED-CONTROL, in response to the statement “What device would you like to control with a smart ring?” for which we provided several options—lights, security system,

⁴<https://www.youtube.com/watch?v=HFUXBONuKPU>, <https://www.youtube.com/watch?v=CP3g3434BNs>, and <https://twitter.com/i/status/790922901025349633>.

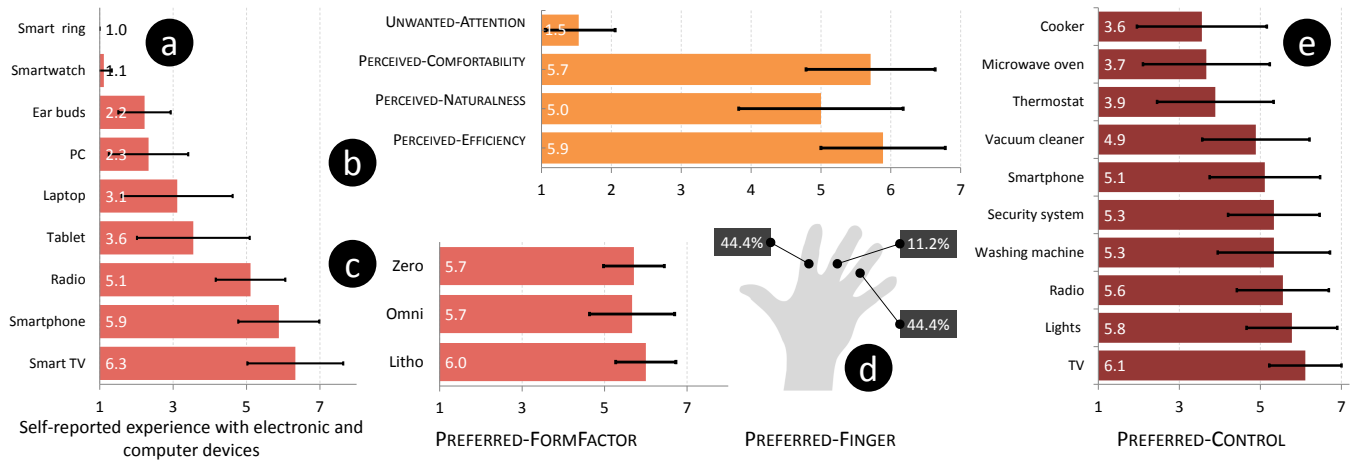


Figure 2: Participants' self-reported experience with smart devices (a), perceptions of smart rings (b), preferences for smart ring form factors (c), fingers on which to wear a smart ring (d), and devices to control with a smart ring (e).

thermostat, washing machine, vacuum cleaner, gas cooker, microwave oven, smart TV, smartphone, radio,—representing common home appliances and electronic devices, but we also encouraged participants to think of other devices they would like to control with a smart ring. For each option, the participants provided a rating between 1 (“not suited at all”) and 7 (“very suited”).

- **PREFERRED-FINGER**, representing the preferred finger on which to wear a smart ring.
- **PREFERRED-FORMFACTOR**, measured with a rating between 1 (“strongly disagree”) and 7 (“strongly agree”) for each of the three smart rings used in our study, in response to the statement “Which of the rings that you have tried (Zero, Litho, and Omni) has the most suitable form factor for you?”

Besides the above specific measures, we also administered two established questionnaires to evaluate our participants' perceptions of smart ring technology with respect to the baseline condition of a *smartphone*, a device used regularly by all of our participants, according to their self-reported experience with smart devices:

- The System Usability Scale (SUS) [7], consisting of ten 5-point Likert statements with items ranging from 1 (“strongly disagree”) to 5 (“strongly agree”). The SUS score varies from 0 (low usability) to 100 (perfect).
- The Technology Acceptance Model (TAM) [10], consisting of twelve 7-point Likert statements with items from 1 (“strongly disagree”) to 7 (“strongly agree”). Based on TAM, we report **PERCEIVED-USEFULNESS** (i.e., “the degree to which a person believes that using a particular system would enhance his or her job performance” [10, p. 320]) and **PERCEIVED-EASEOFUSE** (i.e., “the degree to which a person believes that using a particular system would be free of effort” [10, p. 320]), two fundamental determinants of user acceptance of computer technology. Following [21], we normalized **PERCEIVED-USEFULNESS** and **PERCEIVED-EASEOFUSE** scores in [0,100].

4 RESULTS

We report results about our participants' experience with, perceptions of, and preferences for smart ring technology. Figure 2 gives an overview of all of our empirical findings.

4.1 User Experience

The participants were overall positive about their experience with the smart ring models they tried out during the study, for which they rated **PERCEIVED-EFFICIENCY** at an average of 5.9 (SD=1.4, Mdn=6.0), **PERCEIVED-NATURALNESS** at 5.0 (SD=1.8, Mdn=5.0), and **PERCEIVED-COMFORTABILITY** at 5.7 (SD=1.4, Mdn=5.5) on scales from 1 (low) to 7 (high). Of all of the individual ratings collected for these measures, 77.8% were higher than 5. Furthermore, the perception of **UNWANTED-ATTENTION** drawn to oneself by wearing and using a smart ring in public places was rated low (M=1.5, SD=0.8, Mdn=1), with 66.7% of the individual ratings being below 2. We found a statistically significant positive linear correlation between **PERCEIVED-EFFICIENCY** and **PERCEIVED-NATURALNESS** (Pearson's $r_{(N=9)}=.712$, $CI_{95\%}=[.462,1.000]$).

4.2 Preferences for Smart Rings

The participants preferred, in decreasing order, the form factor of Litho (M=6.0, SD=1.1, Mdn=6.0), Omni (M=5.7, SD=1.6, Mdn=6.0), and Zero (M=4.4, SD=1.1, Mdn=4.0), although a Friedman test did not detect a statistically significant effect of the smart ring model on **PREFERRED-FORMFACTOR** ($\chi^2_{(2)}=3.818$, $p=.158$, $n.s.$). We also found that our participants' choices were split regarding the **PREFERRED-FINGER** on which to wear a smart ring, with four participants (4/9=44.4%) preferring the index finger and other four (44.4%) the ring finger. The finding about the index finger is in line with the results reported by Vatavu and Bilius [41] from a systematic literature review of smart ring gestures. The preference for wearing a smart ring on the ring finger can be explained by the associations explicitly articulated by our participants between a smart ring and a wedding ring, for which we present more details in Subsection 4.5.

We also elicited participants' preferences for possible uses of smart rings; see Figure 2e. The top-3 preferences were, in decreasing order, TV ($M=6.1$, $SD=1.4$, $Mdn=7.0$), lights ($M=5.8$, $SD=1.7$, $Mdn=7.0$), and radio ($M=5.6$, $SD=1.7$, $Mdn=6.0$) on a scale from 1 (low) to 7 (high). The least preferred appliances were the gas cooker ($M=3.6$, $SD=2.5$, $Mdn=3.0$), microwave oven ($M=3.7$, $SD=2.4$, $Mdn=3.0$), and thermostat ($M=3.9$, $SD=2.2$, $Mdn=3.0$). A Friedman test detected a statistically significant effect of device type on PREFERRED-CONTROL ($\chi^2_{(9)}=28.969$, $p<.001$), but Wilcoxon signed-rank tests (Bonferroni-corrected $\alpha=.05/3=.0167$) did not detect significant differences ($p>.083$, $n.s.$) between the top-3 devices.

4.3 Perceived Usability

The mean SUS scores were 60.6 ($SD=9.8$) for the *smart ring* and 70.8 ($SD=11.1$) for the *smartphone* condition, falling under the "marginal" range and between the "OK" and "good" levels of usability, according to Bangor *et al.*'s [1] acceptability ranges and adjective ratings for interpreting SUS scores. These results are explained by our participants' need for assistance on how to use a smart ring (and, in the past, a smartphone), reflected in their answers to the statements "I think that I would need the support of a technical person to be able to use this system." ($M=4.7$ for *smart ring* and $M=3.9$ for *smartphone*) and "I needed to learn a lot of things before I could get going with this system" ($M=4.6$ and 3.9) from the SUS questionnaire. A Wilcoxon signed-rank test revealed a statistically significant difference ($Z = -2.252$, $p=.031$) between the perceived usability of the *smart ring* and *smartphone*, respectively.

4.4 Acceptance of Smart Ring Technology

The mean TAM scores for *smart ring* were 73.8 ($SD=27.1$) in terms of PERCEIVED-USEFULNESS and 76.9 ($SD=21.6$) in terms of PERCEIVED-EASEOFUSE, lower than those obtained in the *smartphone* condition, 80.6 ($SD=8.6$) and 81.2 ($SD=8.9$), but Friedman tests did not detect statistically significant differences between the *smart ring* and *smartphone* on either PERCEIVED-USEFULNESS ($Z = -.356$, $p=.758$, $n.s.$) or PERCEIVED-EASEOFUSE ($Z = .119$, $p=.934$, $n.s.$). We found significant positive linear correlations between SUS and PERCEIVED-USEFULNESS ($r_{(N=9)}=.826$, $CI_{95\%}=[.670, .990]$) and SUS and the PERCEIVED-EASEOFUSE ($r_{(N=9)}=.739$, $CI_{95\%}=[.439, .979]$) of smart rings, respectively.

4.5 Qualitative Feedback

To complement the findings obtained using our quantitative measures, we report in the following participants' comments, which we organize thematically according to the perceived comfort and convenience of smart rings.

4.5.1 Material, form factor, and wearable comfort. Most of the participants preferred the Omni smart ring due to the material from which it is made, a scratch-resistant zirconia ceramic, e.g., P₅ said that "it [the material] looks very good and seems more resistant." Some of the participants considered that the optimal form factor of a smart ring should be similar to that of a wedding ring. For this reason, Zero and Omni were preferred by some of the participants, e.g., "I prefer Omni, it has the thickness of my wedding ring" (P₂), "like Omni, Zero is easy to handle, thin, and doesn't bother me. It is

the same as a wedding ring" (P₄), and "I prefer Zero, although Omni is also easy to use. But I prefer Zero because it is thicker and easier to handle. I can easily notice if I lost it" (P₃). In contrast, Litho's form factor, representing a *ring-like* device, attracted our participants' attention the most. For example, P₉ said "like Omni, it [Litho] has a nice appearance, although I prefer Litho because of its shape. I think it would be nice if Litho were made of the same material as Omni. Litho is more special, it is not a common ring, and has caught my attention," and P₇ mentioned that "I like Litho. It is the most suited for me in terms of thickness and shape. It attracts attention because it is atypical and more special." Other aspects of comfortability were equally highlighted for the three smart ring models. For example, P₅ said "I like Omni, it is thin and easy to wear," P₆ mentioned that "just like Omni, Zero is light and I would enjoy wearing a ring having these dimensions," while P₈ commented about Litho being "smooth to the touch and does not slip" with "perfect thickness so that I can feel it on my finger, Omni and Zero seem too thin." These comments reflect aspects of an aesthetic experience [11] created by the various smart ring models used in our study, e.g., the enjoyment produced by the fine, smooth material of Omni or the intriguing form factor of Litho, representing sensory pleasure occurring at the visceral level of emotional product design [29].

4.5.2 Convenience of wearing a smart ring to control other devices.

Our participants considered the capability of a smart ring to control remote devices useful and convenient. For example, P₈ said that he would use a smart ring "all of the time, especially if the ring would tell me my body temperature and heartbeat. I could lie in bed and turn on the vacuum cleaner or the lights without even getting up." P₇ said she would prefer a smart ring to control the intercom without relying on the access card, which she described as cumbersome: "I would like the ring to replace my intercom card, it would be easier for me to enter the building without having to search for the card in my purse. When I'm at home, I'd like to be able to answer the intercom using the ring." P₉ commented on the convenience of a smart ring with respect to a smartphone: "With my smartphone, I can make and receive calls and go online, but a smart ring is more convenient because I could control devices from the house. Besides, it doesn't bother me. I don't have to take care of it because it is on my finger. I could wear it all of the time, just like my wedding ring. As I have been taking care of my wedding ring for almost forty years now, I can also see myself wearing a smart ring with which to control things." This comment reveals the convenience of wearing a smart ring in everyday life, an aspect of emotional experience [11] resulting from the satisfaction experienced by P₉ that the smart ring matches his needs. From this perspective of convenience, the smart ring is placed in a direct association with the familiarity of wearing the wedding ring. However, the latter also creates an experience of meaning [11],—i.e., the personal, symbolic significance of the wedding ring with profound and sustained meaning to the wearer,—different from the aesthetic and emotional experience discussed previously. The convenience of a smart ring also led to several comments about incentives for purchasing one: P₅ considered that "The ring must bring something new than what a smartphone can already do," P₆ witnessed that "I'll buy one if I knew I could easily control the TV or lights without keep getting out of bed, because my legs hurt and it's hard to get up," while

P₈ emphasized her openness to new technology, “I always want something new. I like it [new technology] a lot.”

5 DISCUSSION

Based on our empirical findings, we propose implications for future designs of smart ring-based interactions and applications for older people by following the four items of Vines *et al.*'s [42] agenda for aging research in HCI.

5.1 Challenge assumptions and predominantly negative societal attitudes of aging

Drawing from previous research on older people's adoption of mobile and wearable devices, the multitude of features offered by new devices [23,47] and the lack of knowledge and confidence [39] are factors limiting their adoption. Despite such stereotypes, common in the scientific literature [42], we found that our participants were open to smart rings as a useful technology in their everyday life. These results corroborate Mitzner *et al.* [26], who reported that older people embrace a diversity of technologies, but large-scale adoption requires adequate knowledge of the benefits. Thus, we believe that user age is not a factor in the adoption of smart rings, but the perceived usefulness of rings is. Future work should examine usage scenarios and applications of smart rings more closely, including specific contexts of use (e.g., at home vs. public places or when standing, walking, or inside a vehicle, etc.) to identify new needs and smart ring features that bring value to users in context.

5.2 The impact of users' personal histories upon technology use now and in the future

According to this agenda item [42], technology design should follow deepened engagement with the users' life course, instead of responding to generalized insights about the capabilities of the larger user group. Thus, it is useful to consider how smart rings can integrate older people's everyday life, now and in the future, in ways that adapt to changing needs [8,39] during one's life course. Most of our participants' comments reflected practical uses of smart rings in everyday life, either for personal health or interacting easier with home appliances and electronic devices, but longitudinal, in-situ studies [3] are recommended to gain a better understanding of motivations and needs of smart ring usage in relation to everyday life as well as expectations of smart ring technology for the future.

5.3 Participatory design with measures of success meaningful to older adults

Our findings revealed various incentives for purchasing and using a smart ring, such as more effective and comfortable interactions for performing control tasks compared to a smartphone. They also revealed analogies with the wedding ring, the pleasure of wearing a well-crafted piece of jewelry, and the desire to learn about new technology in retirement, which also represent measures of success in later life [42, p. 2:21], more specific and varied than successful task completion. Interesting future work could examine more closely such specific nuances of success meaningful to older people using smart rings. Also, participatory design in the form of end-user elicitation studies [44] is recommended to unveil older adults'

preferences for intuitive ring-based interactions and complement findings obtained with young adults [13,15].

5.4 Innovations in interactive technology shaped by older people

This agenda item [42] highlights older people as a resource of ideas and creativity, but also opportunities for them to change dominant narratives of aging in society. Regarding the latter, design knowledge of smart ring technology from the scientific literature has been mostly informed by studying young adult users—which does not reflect accurately the experience of all age groups [33],—while applications for older people have primarily addressed health and/or rehabilitation, one stereotype of aging research [42]. To give older adults a voice in shaping innovations in smart wearables that match their needs and expectations, combining participatory design with the DIY culture, e.g., older people creating customized devices for themselves [18], is an interesting avenue to explore in future work. The benefits are represented by new device form factors, interfaces, features, and functionalities better suited to one's needs, abilities, and preferences [20]. In this regard, participatory design of *ring-ready* devices [41], which are multiform, multi-purpose wearables that can be worn in many ways, including as a ring, represents interesting future work.

6 LIMITATIONS

Our study has several limitations. The number of participants that we recruited is small, and future work is recommended to collect and examine perceptions of and preferences for smart rings using a larger sample of older people. Also, we elicited participants' feedback following just a brief session where the participants tried out several smart rings, whereas more accurate findings will be revealed from studies involving longer usage periods, including of actual smart ring applications, e.g., controlling the TV or making POS payments. Lastly, we used only a small selection of smart ring models in our study, while many other models are commercially available, featuring different form factors, materials, and weights. Further studies are recommended to complete our findings.

7 CONCLUSION

We reported empirical findings about older people's perceptions of and preferences for smart ring technology. Overall, we found a generally positive view toward adopting smart rings due to their perceived usefulness in everyday life. We also presented future work opportunities for designing smart ring technology for older people in accordance with the aging research agenda in HCI. We look forward to more future work in this direction.

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